

2. Apply data pre-processing techniques such as standardization/normalization, transformation, aggregation, discretization/binarization, sampling etc. on any dataset

```
In [1]: import pandas as pd
import numpy as np
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler, MinMaxScaler, Binarizer

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = pd.Categorical.from_codes(iris.target, iris.target_names)

print(df.head())
print("\nDataset shape:", df.shape)
print("\nSummary statistics:\n", df.describe())

X = df.drop(columns=['species'])
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

	species
0	setosa
1	setosa
2	setosa
3	setosa
4	setosa

Dataset shape: (150, 5)

Summary statistics:

	sepal length (cm)	sepal width (cm)	petal length (cm)	\
count	150.000000	150.000000	150.000000	
mean	5.843333	3.057333	3.758000	
std	0.828066	0.435866	1.765298	
min	4.300000	2.000000	1.000000	
25%	5.100000	2.800000	1.600000	
50%	5.800000	3.000000	4.350000	
75%	6.400000	3.300000	5.100000	
max	7.900000	4.400000	6.900000	

	petal width (cm)
count	150.000000
mean	1.199333
std	0.762238
min	0.100000
25%	0.300000
50%	1.300000
75%	1.800000
max	2.500000

```
In [2]: scaler = StandardScaler()
df_standardized = scaler.fit_transform(X)
df_standardized = pd.DataFrame(df_standardized, columns=X.columns)

print("\nStandardized Data (first 5 rows):\n", df_standardized.head())
```

Standardized Data (first 5 rows):

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	-0.900681	1.019004	-1.340227	-1.315444
1	-1.143017	-0.131979	-1.340227	-1.315444
2	-1.385353	0.328414	-1.397064	-1.315444
3	-1.506521	0.098217	-1.283389	-1.315444
4	-1.021849	1.249201	-1.340227	-1.315444

```
In [3]: normalizer = MinMaxScaler()
df_normalized = normalizer.fit_transform(X)
df_normalized = pd.DataFrame(df_normalized, columns=X.columns)

print("\nNormalized Data (first 5 rows):\n", df_normalized.head())
```

Normalized Data (first 5 rows):

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	0.222222	0.625000	0.067797	0.041667
1	0.166667	0.416667	0.067797	0.041667
2	0.111111	0.500000	0.050847	0.041667
3	0.083333	0.458333	0.084746	0.041667
4	0.194444	0.666667	0.067797	0.041667

```
In [4]: df_transformed = df.copy()
df_transformed["sepal width (cm)"] = np.log1p(df_transformed["sepal width (cm)"])

print("\nTransformed (log) Data (first 5 rows):\n", df_transformed.head())
```

Transformed (log) Data (first 5 rows):

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	1.504077	1.4	0.2	
1	4.9	1.386294	1.4	0.2	
2	4.7	1.435085	1.3	0.2	
3	4.6	1.410987	1.5	0.2	
4	5.0	1.526056	1.4	0.2	

	species
0	setosa
1	setosa
2	setosa
3	setosa
4	setosa

```
In [5]: agg_df = df.groupby("species", observed=False).mean()
print("\nAggregated Mean by Species:\n", agg_df)
```

Aggregated Mean by Species:

	sepal length (cm)	sepal width (cm)	petal length (cm)	\
species				
setosa	5.006	3.428	1.462	
versicolor	5.936	2.770	4.260	
virginica	6.588	2.974	5.552	

	petal width (cm)
species	
setosa	0.246
versicolor	1.326
virginica	2.026

```
In [6]: bins = [0, 2.5, 5, 7.5]
labels = ['Small', 'Medium', 'Large']

df_discrete = df.copy()
df_discrete['custom_petal_length'] = pd.cut(df['petal length (cm)'], bins=bins, labels=labels)

print("\nDiscretized Data (Petal Length into 3 bins):\n", df_discrete.head())
```

Discretized Data (Petal Length into 3 bins):

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

	species	custom_petal_length
0	setosa	Small
1	setosa	Small
2	setosa	Small
3	setosa	Small
4	setosa	Small

```
In [7]: binarizer = Binarizer(threshold=1.0)
df_binarized = df.copy()
df_binarized["wide_petal"] = binarizer.fit_transform(df[["petal width (cm)"]])

print("\nBinarized Data (Petal width >= 1.0 -> 1):\n", df_binarized.head())
```

Binarized Data (Petal width >= 1.0 -> 1):

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

	species	wide_petal
0	setosa	0.0
1	setosa	0.0
2	setosa	0.0
3	setosa	0.0
4	setosa	0.0

```
In [8]: sample_df = df.sample(frac=0.2, random_state=42)
print("\nSampled 20% of the Data:\n", sample_df)
```

Sampled 20% of the Data:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
73	6.1	2.8	4.7	1.2	
18	5.7	3.8	1.7	0.3	
118	7.7	2.6	6.9	2.3	
78	6.0	2.9	4.5	1.5	
76	6.8	2.8	4.8	1.4	
31	5.4	3.4	1.5	0.4	
64	5.6	2.9	3.6	1.3	
141	6.9	3.1	5.1	2.3	
68	6.2	2.2	4.5	1.5	
82	5.8	2.7	3.9	1.2	
110	6.5	3.2	5.1	2.0	
12	4.8	3.0	1.4	0.1	
36	5.5	3.5	1.3	0.2	
9	4.9	3.1	1.5	0.1	
19	5.1	3.8	1.5	0.3	
56	6.3	3.3	4.7	1.6	
104	6.5	3.0	5.8	2.2	
69	5.6	2.5	3.9	1.1	
55	5.7	2.8	4.5	1.3	
132	6.4	2.8	5.6	2.2	
29	4.7	3.2	1.6	0.2	
127	6.1	3.0	4.9	1.8	
26	5.0	3.4	1.6	0.4	
128	6.4	2.8	5.6	2.1	
131	7.9	3.8	6.4	2.0	
145	6.7	3.0	5.2	2.3	
108	6.7	2.5	5.8	1.8	
143	6.8	3.2	5.9	2.3	
45	4.8	3.0	1.4	0.3	
30	4.8	3.1	1.6	0.2	

	species
73	versicolor
18	setosa
118	virginica
78	versicolor
76	versicolor
31	setosa
64	versicolor
141	virginica
68	versicolor
82	versicolor
110	virginica
12	setosa
36	setosa
9	setosa
19	setosa
56	versicolor
104	virginica
69	versicolor
55	versicolor
132	virginica
29	setosa
127	virginica

26	setosa
128	virginica
131	virginica
145	virginica
108	virginica
143	virginica
45	setosa
30	setosa